

Project Song-Reco:

Hardware-Assisted Song Recognition

A Cross-Lingual Beat & Lyric Search Engine Interface

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Note

Welcome to **Song-Reco**. To be clear: the name stands for Song *Recognition*, not Recommendation! Our goal is to identify exactly what song is playing using a standard Arduino starter kit. Because microcontrollers lack the memory to store large databases or process complex audio frequencies, this project will teach you **Edge Computing**. You will use the Arduino to capture physical data (the rhythm/beats of the song via a sound sensor or button), and pass that data via Serial to your PC. Your PC will handle the heavy lifting, searching the rhythm or lyrics in English, Hindi (हिन्दी), नेपाली (नेपाली), and 한국어 (한국어). Let's get to work.

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1 Background Information & Theory

How does a computer "listen" to a song and know its name? Commercial apps like Shazam use **Audio Fingerprinting**. They map the peak frequencies of a song over time to create a unique digital "constellation" or hash.

However, standard Arduino kits only contain **KY-037 or KY-038 Sound Sensors**. These cannot measure specific frequencies (pitch); they can only measure **volume amplitude** (loudness).

1.1 The Song-Reco Strategy

Since we cannot fingerprint frequencies, we will identify songs using two parallel methods:

1. **Beat/Rhythm Hashing:** The user holds the Arduino near the speaker. The sound sensor detects loud volume spikes (the drum beats). The Arduino records the milliseconds between these beats (the tempo/rhythm signature) and sends this array to the PC.
2. **Lyric Search Input:** The user can use the hardware interface to trigger a "listening mode" where they type known lyrics into the serial terminal, allowing the backend to search cross-lingual databases.

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2 Hardware Bill of Materials (BOM)

Gather the following components from your Liberty College standard kits:

- **1x Arduino Uno R3**
- **1x KY-037 Sound Sensor Module** (Microphone with analog & digital output)
- **1x Push Button** (To trigger "Record" mode)
- **3x LEDs** (Red = Standby, Yellow = Recording/Listening, Green = Match Found)
- **3x 220Ω Resistors** (For the LEDs)
- **Breadboard and Jumper Wires**

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3 Arduino Implementation Guide

3.1 Circuit Wiring

1. Connect the **A0 pin** on the KY-037 Sound Sensor to **Analog Pin A0** on the Arduino.
2. Connect the sensor's **GND** and **VCC** to the Arduino's 5V and GND.
3. Wire the **Push Button** from Pin 2 to GND (we will use internal pull-up logic).
4. Wire the **Red LED** to Pin 8, **Yellow LED** to Pin 9, and **Green LED** to Pin 10 (with 220Ω resistors connecting them to GND).

3.2 Phase 1 Firmware (Beat Extraction Core)

This C++ code waits for the user to press the button, then records the timing of volume peaks (drum beats) for 10 seconds. It packages this rhythm data and sends it over the Serial port.

```
1  /*
2   Project: Song-Reco
3   Supervisor: Danish Puri, Liberty College
4   Description: Hardware interface to capture rhythmic beat signatures.
5  */
6
7  const int micPin = A0;          // KY-037 Analog Out
8  const int recordBtn = 2;        // Start recording button
9  const int ledRed = 8;           // Standby
10 const int ledYellow = 9;         // Recording
11 const int ledGreen = 10;         // Success
12
13 int threshold = 600;             // Adjust based on room noise (0-1023)
14 bool isRecording = false;
15 unsigned long recordStartTime;
16
17 void setup() {
18   Serial.begin(9600);
19   pinMode(recordBtn, INPUT_PULLUP);
20   pinMode(ledRed, OUTPUT);
21   pinMode(ledYellow, OUTPUT);
22   pinMode(ledGreen, OUTPUT);
23
24   digitalWrite(ledRed, HIGH); // System Standby
25   Serial.println("SONG-RECO SYSTEM ONLINE. Awaiting command.");
26 }
27
28 void loop() {
29   // Check PC Serial for "SUCCESS" message to light green LED
30   if (Serial.available() > 0) {
31     String incoming = Serial.readStringUntil('\n');
32     if (incoming == "MATCH_FOUND") {
33       digitalWrite(ledRed, LOW);
34       digitalWrite(ledGreen, HIGH);
35       delay(5000); // Show success for 5 seconds
36       digitalWrite(ledGreen, LOW);
37       digitalWrite(ledRed, HIGH);
38     }
39   }
40
41   // Start Recording sequence
42   if (digitalRead(recordBtn) == LOW && !isRecording) {
43     isRecording = true;
44     digitalWrite(ledRed, LOW);
45     digitalWrite(ledYellow, HIGH);
46     Serial.println("RECORDING_START");
```

```

47   recordStartTime = millis();
48   delay(200); // Debounce
49 }
50
51 // While recording, detect loud beats
52 if (isRecording) {
53     int micValue = analogRead(micPin);
54
55     if (micValue > threshold) {
56         // Send the timestamp of the beat to the PC
57         Serial.print("BEAT:");
58         Serial.println(millis() - recordStartTime);
59         delay(150); // Prevent double-counting the same beat pulse
60     }
61
62     // Stop recording after 10 seconds (10000 ms)
63     if (millis() - recordStartTime > 10000) {
64         isRecording = false;
65         digitalWrite(ledYellow, LOW);
66         digitalWrite(ledRed, HIGH);
67         Serial.println("RECORDING_END");
68     }
69 }
70 }

```

Listing 1: Song-Reco Hardware Interface (SongReco.ino)

4 Phase 2: The PC Software Target

While writing the PC software is the next phase of the project, students should conceptualize the Python script that will catch the Arduino's data.

When the Arduino sends data like:

```

RECORDING_START
BEAT:450
BEAT:900
BEAT:1350
RECORDING_END

```

The Python script (using the `pyserial` library) will calculate the gaps between beats (450ms → roughly 133 Beats Per Minute). It will then query an API or a local JSON database containing tempos and lyrics.

If the user typed a lyric query into the Python interface simultaneously the Python script cross-references the lyrics and the Arduino's detected tempo to return the exact song match!